

SUBNETTING MASTERY AND MOCK EXAM

1. Given the network 192.168.10.0/24, divide it into 4 equal subnets and write down the network address, first usable IP, last usable IP, and broadcast address for each subnet.

->**Divide 192.168.10.0/24 into 4 equal subnets**

A /24 has 256 addresses. To create **4 equal subnets**, borrow 2 bits:

/24 → /26

Subnet mask: **255.255.255.192**

Each /26 has 64 total addresses and **62 usable addresses**.

Subnet	Network Address	First Usable	Last Usable	Broadcast
1	192.168.10.0/26	192.168.10.1	192.168.10.62	192.168.10.63
2	192.168.10.64/26	192.168.10.65	192.168.10.126	192.168.10.127
3	192.168.10.128/26	192.168.10.129	192.168.10.190	192.168.10.191
4	192.168.10.192/26	192.168.10.193	192.168.10.254	192.168.10.255

2. You are setting up a Zomato-style food delivery office with 3 departments: Customer Support (50 devices), Delivery Management (30 devices), and Marketing (20 devices). Design a subnetting plan using the 10.0.0.0/24 network so that each department has its own subnet, maximizing available addresses and minimizing waste. List the subnet mask and IP range for each department.

->**Zomato-style food delivery office — VLSM**

Base network: **10.0.0.0/24**

Requirements:

- Customer Support → 50 devices

- Delivery Management → 30 devices
- Marketing → 20 devices

Using VLSM, allocate the largest requirement first.

Department	Devices	CIDR	Subnet Mask	Network Address	Usable IP Range	Broadcast
Customer Support	50	/26	255.255.255.192	10.0.0.0	10.0.0.1 – 10.0.0.62	10.0.0.63
Delivery Management	30	/27	255.255.255.224	10.0.0.64	10.0.0.65 – 10.0.0.94	10.0.0.95
Marketing	20	/27	255.255.255.224	10.0.0.96	10.0.0.97 – 10.0.0.126	10.0.0.127

Why?

- /26 → 62 usable addresses → enough for 50 devices.
 - /27 → 30 usable addresses → exactly enough for 30 devices.
 - /27 → 30 usable addresses → enough for 20 devices.
3. Troubleshoot the following scenario: Two teams in your Flipkart warehouse are accidentally assigned overlapping subnets (Team A: 172.16.10.0/25, Team B: 172.16.10.64/26). Identify the overlapping IP range and suggest a corrected subnet plan so both teams have unique IP ranges.

Hint: Draw the address ranges for both subnets to spot the overlap.

-> **Overlapping Flipkart warehouse subnets**

Given:

Team A: 172.16.10.0/25

A /25 has 128 addresses:

Network: 172.16.10.0

Usable: 172.16.10.1 – 172.16.10.126

Broadcast: 172.16.10.127

Team B: 172.16.10.64/26

A /26 has 64 addresses:

Network: 172.16.10.64

Usable: 172.16.10.65 – 172.16.10.126

Broadcast: 172.16.10.127

Overlapping range

The overlapping address range is:

172.16.10.64 – 172.16.10.127

The usable IP overlap is:

172.16.10.65 – 172.16.10.126

Corrected plan

Keep Team A as:

Team A: 172.16.10.0/25

Usable: 172.16.10.1 – 172.16.10.126

Move Team B to the next available /26:

Team B: 172.16.10.128/26

Usable: 172.16.10.129 – 172.16.10.190

Broadcast: 172.16.10.191

Now the two teams have **completely separate IP ranges**.

4. You are managing a growing IPL Fantasy League app backend and notice that your 192.168.1.0/26 subnet is running out of IPs. Propose a solution to handle address exhaustion so new servers can be added without disrupting existing users.

Hint: Consider subnet resizing or migrating to a larger subnet.

-> **IPL Fantasy League backend running out of IPs**

Current subnet:

192.168.1.0/26

A /26 provides:

- 64 total addresses
- **62 usable addresses**
- Usable range: 192.168.1.1 – 192.168.1.62
- Broadcast: 192.168.1.63

If more servers are required, the subnet needs to be enlarged.

Recommended solution: migrate to a /25

Use:

192.168.1.0/25

This provides:

- 128 total addresses
- **126 usable addresses**
- Usable range: 192.168.1.1 – 192.168.1.126
- Broadcast: 192.168.1.127

This doubles the available usable addresses from **62 to 126**.

Important consideration

Simply changing /26 to /25 isn't always possible without planning, because the larger subnet changes the address boundary and routing configuration.

A practical migration would be:

1. Plan the new /25 subnet.
2. Verify that 192.168.1.64–192.168.1.126 isn't being used by another network.
3. Update routing/DHCP/firewall configurations.
4. Migrate servers gradually.
5. Keep existing addresses where possible to minimize disruption.
6. Test connectivity before removing the old configuration.

For a production application, another option is to allocate a **new larger subnet** and migrate servers to it gradually, avoiding an abrupt change.

5. After completing your subnetting speed round and mock exam, review your answers and identify 3 types of mistakes you made (e.g., miscalculating subnet masks, overlapping ranges, wrong broadcast address). For each, write how you would avoid the mistake next time.

->**Mistake 1: Miscalculating the subnet mask**

Problem: Confusing /25, /26, and /27.

How to avoid it: Memorize the common masks:

/24 = 255.255.255.0

/25 = 255.255.255.128

/26 = 255.255.255.192

/27 = 255.255.255.224

/28 = 255.255.255.240

Also remember the usable-host formula:

Usable hosts = $2^{\text{host bits}} - 2$

Mistake 2: Creating overlapping subnets

Problem: Assigning a subnet that falls inside another subnet's address range.

How to avoid it: Always write down:

Network → Usable Range → Broadcast

before assigning the next subnet.

For example:

172.16.10.0/25

↓

0 – 127

So another subnet cannot start at .64.

Mistake 3: Calculating the wrong broadcast address

Problem: Treating the last usable IP as the broadcast address.

How to avoid it: Remember:

Broadcast = last address in the subnet

For example:

192.168.10.64/26

Network: .64

Usable: .65 – .126

Broadcast: .127

A quick method is to calculate the **block size**:

$256 - 192 = 64$

So /26 networks begin at:

0, 64, 128, 192

This makes network and broadcast calculations much faster.